

Impact of COVID-19 on JSE/ALSI Market Volatility

Statistical analysis of absolute daily returns across Pre-COVID, COVID Shock and Recovery periods

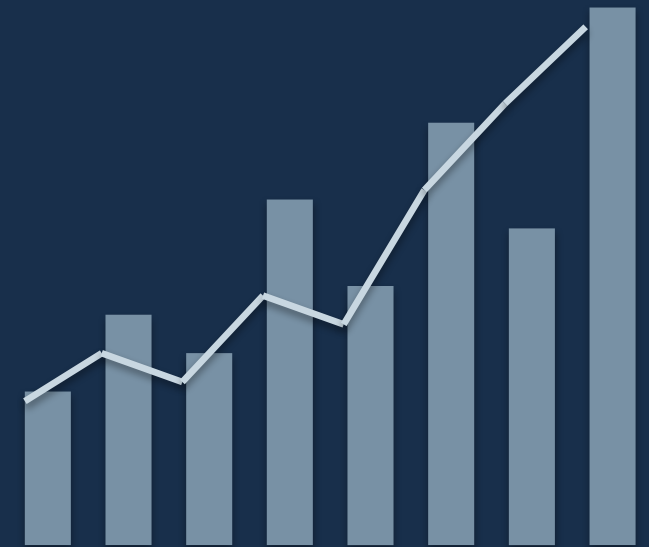
Computational Finance Project

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Executive Summary

What the statistical analysis shows

TOTAL OBSERVATIONS

501

02 Jan 2019–31 Dec 2020 dataset range*

ANALYTICAL OBSERVATIONS

252

Across three defined periods

ANOVA

F = 57.96

p < 0.001

KRUSKAL-WALLIS

$\chi^2 = 23.634$

p = 0.0001

Absolute daily returns were substantially higher during the COVID Shock period (mean = 3.9949%) than Pre-COVID (0.9104%) and Recovery (1.0200%).

Shapiro–Wilk testing rejects normality for Pre-COVID and Recovery, while the COVID Shock sample does not reject normality (p = 0.28888).

The robust variance test and Bartlett’s test both indicate significant differences in group variances (p < 0.001).

The Kruskal–Wallis test confirms a statistically significant difference across the three periods.

Pairwise Wilcoxon tests show significant differences for Pre-COVID vs COVID Shock and COVID Shock vs Recovery, but not for Pre-COVID vs Recovery.

*The Stata date summary covers 02 Jan 2019 to 31 Dec 2020; 249 observations fall outside the three analytical period labels.

Research Objective & Hypotheses

From market disruption to statistical inference

PRIMARY OBJECTIVE

Assess whether volatility changed

Compare absolute daily returns across three COVID-related periods

Research questions

Did the magnitude of daily market movements differ across the three periods?

Was the COVID Shock period associated with materially higher absolute returns?

Did volatility return to a level statistically comparable with Pre-COVID conditions during Recovery?

Statistical hypotheses

H_0 : The distributions / mean absolute daily returns are equal across the three periods.

H_1 : At least one period differs from the others.

Pairwise H_0 : Each specified pair has the same distribution of absolute daily returns.

For the three pairwise tests, Bonferroni-adjusted $\alpha = 0.05/3 = 0.01667$.

Data & Study Design

JSE/ALSI daily market observations and analytical segmentation

DATASET

501 rows

Final Stata dataset

DATE RANGE

02 Jan 2019

to 31 Dec 2020

RETURN SERIES

500 valid

1 expected first return missing

ANALYSIS SAMPLE

252

45 + 19 + 188

Core variables

- ``date_stata`` — Stata-formatted trading date
- ``close`` — closing market level used for return calculation
- ``daily_return`` — percentage daily return
- ``abs_daily_return`` — absolute daily return used as the volatility proxy
- ``period`` — Pre-COVID, COVID Shock and Recovery classification

Period allocation

Period	N	Share of full dataset
Pre-COVID	45	8.98%
COVID Shock	19	3.79%
Recovery	188	37.52%
Outside defined periods	249	49.70%
Analytical total	252	50.30%

Analytical Framework

Why both parametric and non-parametric inference was retained

1

Data validation

Dates, missing values, duplicates

2

Return construction

Daily % return and absolute return

3

Descriptive analysis

N, mean, SD, min, max

4

Assumption testing

Shapiro-Wilk + variance tests

5

Inference

ANOVA + Kruskal-Wallis

6

Pairwise analysis

Wilcoxon rank-sum + Bonferroni

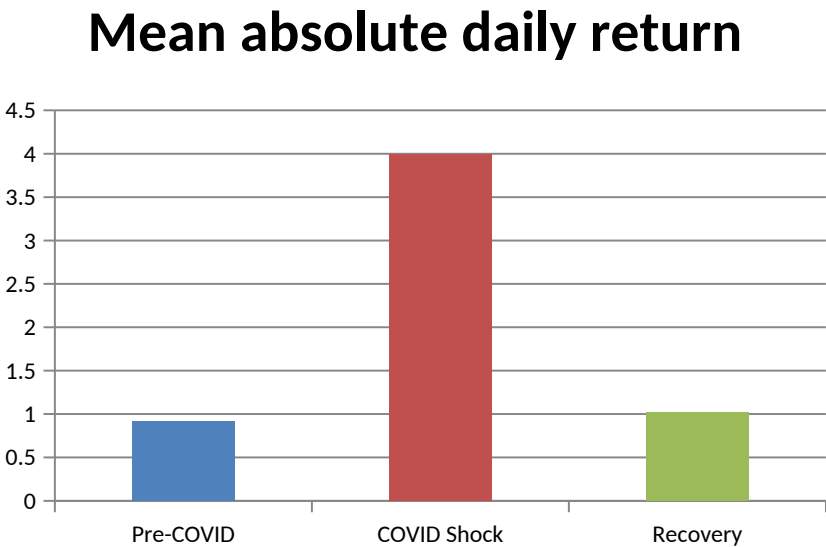
Method choice: ANOVA is reported as the classical mean-comparison test, while Kruskal-Wallis and Wilcoxon provide assumption-robust non-

Return construction: Daily return = $[(\text{Close}_t - \text{Close}_{t-1}) / \text{Close}_{t-1}] \times 100$; volatility proxy = $|\text{Daily return}|$.

Descriptive Statistics

Absolute daily return (%) by analytical period

Period	N	Mean	SD	Min	Max
Pre-COVID	45	0.9104	1.0140	0.0451	4.5031
COVID Shock	19	3.9949	2.9500	0.0180	9.7213
Recovery	188	1.0200	0.8584	0.0038	3.8802
Total	252	1.2247	1.4083	0.0038	9.7213



Key descriptive insight

COVID Shock has the highest mean absolute daily return (3.9949%) and the highest standard deviation (2.9500%).

Recovery mean (1.0200%) is close to Pre-COVID (0.9104%), while dispersion is lower in Recovery (0.8584% vs 1.0140%).

Normality Assessment

Shapiro–Wilk test and Stata Q–Q evidence

Period	N	W	p-value	Decision
Pre-COVID	45	0.75482	<0.001	Reject H_0
COVID Shock	19	0.94221	0.28888	Fail to reject H_0
Recovery	188	0.88473	<0.001	Reject H_0

Interpretation

Normality is rejected for Pre-COVID and Recovery.

COVID Shock does not show significant evidence against normality at the 5% level.

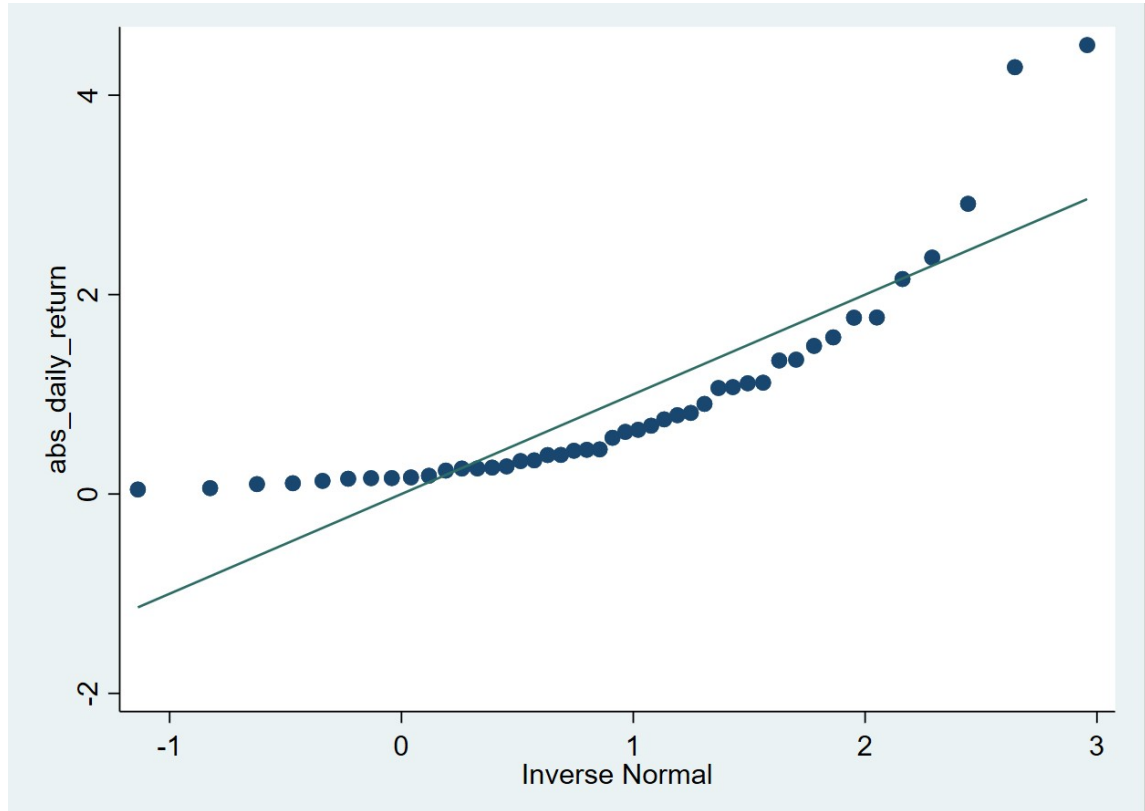
Because normality is not satisfied across all groups, non-parametric confirmation is important.

. bysort period: swilk abs_daily_return					
-> period = Pre-COVID					
Shapiro-Wilk W test for normal data					
Variable	Obs	W	V	z	Prob>z
abs_daily_~n	45	0.75482	10.617	5.007	0.00000
-> period = COVID Shock					
Shapiro-Wilk W test for normal data					
Variable	Obs	W	V	z	Prob>z
abs_daily_~n	19	0.94221	1.319	0.557	0.28888
-> period = Recovery					
Shapiro-Wilk W test for normal data					
Variable	Obs	W	V	z	Prob>z
abs_daily_~n	188	0.88473	16.297	6.401	0.00000

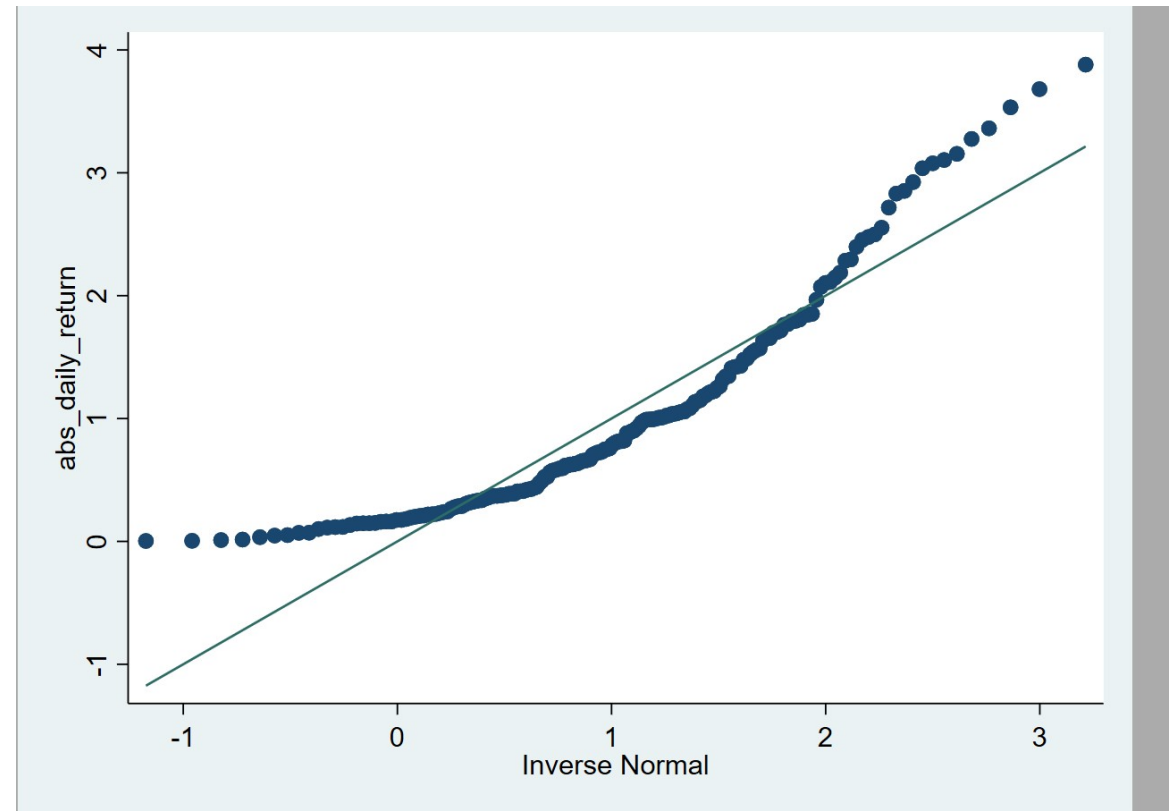
Stata output: Shapiro–Wilk W tests by period

Normality: Q-Q Plots

Graphical assessment supports the formal test results



Stata Q-Q plot shown from the submitted analysis output; visible curvature and upper-tail departures are consistent with non-normality.



Stata Q-Q plot shown from the submitted analysis output; pronounced upper-tail departure is visible.

Equality of Variances

Robust variance testing before interpreting group differences

ROBUST W50 TEST

W50 = 56.7504

F(2,249), p < 0.001

BARTLETT TEST

$\chi^2(2) = 93.0253$

p < 0.001

DECISION

Reject H₀

Variances are not equal

What the result means

The equal-variance assumption is not supported across the three periods.

COVID Shock shows much greater dispersion than the other periods.

This strengthens the case for reporting non-parametric evidence alongside the classical ANOVA.

Group dispersion from descriptive statistics

Period	Mean	SD	Variance (approx.)
Pre-COVID	0.9104	1.0140	1.0283
COVID Shock	3.9949	2.9500	8.7026
Recovery	1.0200	0.8584	0.7369

One-Way ANOVA

Classical comparison of mean absolute daily returns

```
. oneway abs_daily_return period, tabulate
```

period	Summary of abs_daily_return		
	Mean	Std. dev.	Freq.
Pre-COVID	.9103714	1.0140428	45
COVID Sho	3.9949442	2.950014	19
Recovery	1.0199578	.85840449	188
Total	1.2246934	1.4083151	252

Source	Analysis of variance			F	Prob > F
	SS	df	MS		
Between groups	158.137761	2	79.0688803	57.96	0.0000
Within groups	339.683425	249	1.36419046		
Total	497.821185	251	1.98335134		

Bartlett's equal-variances test: chi2(2) = 93.0253 Prob>chi2 = 0.000

ANOVA RESULT

F(2,249) = 57.96

p < 0.001

DECISION

Reject H₀

Mean absolute returns differ

Interpretation

The three group means are not all equal.

The strongest descriptive shift is the elevated COVID Shock mean.

Stata output retained as an evidence snapshot; the presentation table above is used for readability.

Kruskal-Wallis Test

Non-parametric confirmation of differences across all three periods

TEST STATISTIC

$\chi^2(2) = 23.634$

Tie-adjusted result is identical

P-VALUE

0.0001

Statistically significant

DECISION

Reject H₀

At least one period differs

Rank-sum evidence

Period	N	Rank sum
Pre-COVID	45	4,813.00
COVID Shock	19	3,828.00
Recovery	188	23,237.00

Why this matters

The result does not rely on normality or equal-variance assumptions.

It confirms that the distribution/rank location of absolute daily returns differs across the three periods.

This independently supports the central conclusion from ANOVA.

Pairwise Wilcoxon Rank-Sum Tests

Identifying which period comparisons drive the overall difference

Comparison	z	p-value	At $\alpha^*=0.01667$	Decision
Pre-COVID vs COVID Shock	-4.254	<0.001	Significant	Reject H_0
COVID Shock vs Recovery	+4.561	<0.001	Significant	Reject H_0
Pre-COVID vs Recovery	-1.453	0.1463	Not significant	Fail to reject H_0

Interpretation

COVID Shock is significantly different from both Pre-COVID and Recovery.

Pre-COVID and Recovery are statistically comparable in the pairwise rank-sum test.

This pattern is consistent with a temporary shock followed by normalization.

Multiple-comparison control

Three pairwise tests were conducted. The Bonferroni-adjusted threshold was set at $\alpha^* = 0.01667$. The two COVID Shock comparisons remain significant; Pre-COVID vs Recovery does not.

Note: Stata's default two-sided p-values are reported for the rank-sum tests.

Integrated Findings

Putting descriptive, parametric and non-parametric evidence together

The evidence points to a pronounced, statistically significant increase in market-movement magnitude during the COVID Shock period, followed by substantial normalization during Recovery.

MAGNITUDE

COVID Shock mean absolute return = 3.9949%, versus 0.9104% Pre-COVID and 1.0200% Recovery.

ASSUMPTIONS

Normality is not satisfied in two groups and variances are significantly unequal; therefore assumption-robust evidence is important.

INFERENCE

ANOVA and Kruskal-Wallis both reject equality across periods; pairwise tests isolate the COVID Shock differences.

Overall analytical message

The COVID Shock period is the clear outlier in the analysis. Recovery is not statistically different from Pre-COVID in the pairwise Wilcoxon test, suggesting a return toward the pre-shock distribution of absolute daily returns.

Conclusion

Answering the research questions

COVID Shock is associated with a markedly higher magnitude of daily market movements, reflected in the highest mean and dispersion of absolute daily returns.

The difference across periods is statistically significant under both classical ANOVA and the non-parametric Kruskal–Wallis framework.

Pairwise Wilcoxon tests confirm significant differences between COVID Shock and each of the other periods.

The Pre-COVID vs Recovery comparison is not statistically significant ($p = 0.1463$), indicating no evidence of a persistent difference between these two periods under the selected non-parametric test.

The results therefore support a temporary COVID-related disruption followed by substantial recovery toward pre-COVID conditions.

Final conclusion: The JSE/ALSI market exhibited a clear volatility shock during the COVID Shock period, while Recovery moved back toward the Pre-COVID level.

Limitations & Methodological Notes

Important qualifications for interpretation

The analysis uses absolute daily return as a proxy for the magnitude of market movements; it is not a direct estimate of implied or realized volatility.

The three analytical periods are defined classifications within the dataset; 249 of the 501 observations fall outside those period labels and are not used in the inferential comparison.

The COVID Shock group contains only 19 observations, so its normality assessment has limited power and should be interpreted alongside graphical and non-parametric evidence.

The unequal-variance and non-normality findings mean the ANOVA result should be interpreted together with the Kruskal–Wallis and Wilcoxon results rather than in isolation.

The analysis establishes statistical association across periods; it does not by itself identify every causal mechanism behind market movements.

Submission & Reproducibility Files

Final Stata artifacts used for the analysis

FINAL DATA FILE

JSE_ALSI_COVID_FINAL_ANALYSIS.dta

Final Stata dataset with analytical variables

FINAL LOG FILE

JSE_ALSI_COVID_FINAL_ANALYSIS.smcl

Final closed SMCL log containing the analysis

Key variables in the final dataset

date_stata
daily_return
abs_daily_return
period

Key analyses in the final SMCL log

Descriptive statistics
Shapiro–Wilk normality
Robust variance test
ANOVA, Kruskal–Wallis and Wilcoxon tests

Thank You

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